

MASTER'S THESIS RESEARCH SUMMARY

Presenting a Sustainable Closed-Loop Supply Chain Model for Perishable Agri-Food Products with Uncertain Demand (2024)

Author:	Parmis Saboktakin (M.Sc. Graduate) Parmissaboktakin@gmail.com
Institution / Department:	Department of Industrial Engineering / Supply Chain University of Science & Culture
Supervisor(s):	Prof. Rashed Sahraeian sahraeian@shahed.ac.ir
Thesis Grade / Key Impact:	Grade: 19.5/20 (A+)

1. Executive Summary

In the conducted research, the optimization of accumulation and logistical risk management for perishable agricultural products under conditions of uncertain demand has been evaluated. This study develops a multi-period, multi-product mixed-integer linear/nonlinear programming (MILP/MINLP) model for the integrated design of a closed-loop supply chain that simultaneously integrates fleet routing decisions, age-based inventory control, and the reverse flow of waste.

The primary methodological innovation lies in the simultaneous mapping formulation among the deterioration rate, precise calculation of shortage/penalty costs, and waste recovery. Empirical evaluation of the model on the distribution network of the Tehran metropolis indicates that supply chain costs exhibit the highest dependency on seasonal fluctuations, such that peak costs were recorded in summer and the lowest in winter. Furthermore, increasing the shelf-life threshold of goods from 3 to 5 periods during cold seasons significantly increased holding costs while controlling deterioration risk. A 10 percent sensitivity analysis of demand proved the fully linear behavior of the model in response to network workload, with dense regions 4, 5, and 15 accounting for the highest cost shares. The proposed framework provides a dynamic and practical tool for urban logistics decision-making and waste management based on the circular economy.

2. Problem Statement & Research Gap

- **Main Problem:** Optimizing customer responsiveness while reducing total supply chain costs for perishable agri-food products under demand uncertainty.
- **Problem Under Investigation:** Developing a multi-period, multi-product vehicle routing problem (Multi-Period, Multi-Product VRP) that integrates inventory control, product age/perishability considerations, and reverse logistics management (collecting waste and degraded products from distribution centers and returning them to the main warehouse/recycling facilities to transform them into alternative products, such as organic fertilizers).

- **Primary Methodology:** Mathematical modeling via Mixed-Integer Linear/Non-Linear Programming (MILP/MINLP) solved using exact mathematical solvers (e.g., CPLEX/BARON) based on Branch-and-Bound / Branch-and-Cut algorithms.

- **Innovations:**

1. Establishing an integrated closed-loop supply chain (CLSC) framework that simultaneously optimizes three decision-making levels: fleet routing, age-based inventory control, and waste recycling/repurposing.
2. Formulating a synchronized functional relationship between decay rates (b), accurate stockout/penalty and degradation costs, and reverse waste flows from the distribution level back to the supply network.

- **Key Numerical Results:**

1. Seasonal Pattern Effect: Supply chain cost peaks were observed during summer (driven by demand surges and reduced product shelf life), whereas winter yielded the lowest overall network costs.
2. Geographic Cost Distribution: High-density and high-demand zones in the Tehran metropolis (specifically Districts 4, 5, and 15) accounted for the largest share of allocation, warehousing, and routing expenses.
3. Demand Sensitivity Analysis: Applying a 10% fluctuation to demand confirmed a strictly linear response from the model regarding network workload variations.
4. Shelf-Life Analysis (B): Increasing the upper threshold for allowable product age from B=3 to B=5 during colder seasons mitigated degradation risks; however, it noticeably elevated total inventory holding costs due to higher storage volumes at distribution centers.

- **Research Rationale:** Mitigating substantial waste in perishable agricultural produce, optimizing urban logistics costs, and advancing environmental sustainability through circular economy practices (reintegrating waste into the consumption cycle).

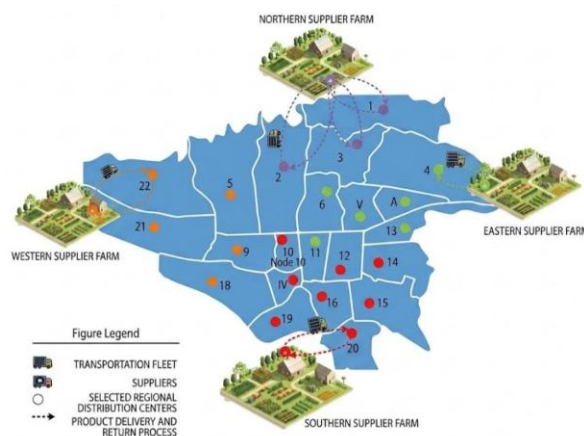


Figure 1-1 Case study of the distribution problem in Tehran city.

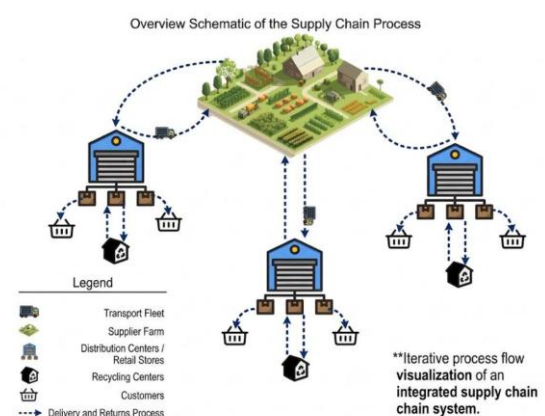


Figure 1-1 – Comprehensive View of the Project Process

3. Methodology & System Framework

Literature Review & Background

Past studies were evaluated based on objective functions, parameter types, chain structure, vehicle routing, and solution approaches to contextualize this research within the literature:

- **Jiang et al. (2022)**: Single-objective model (economic-environmental), deterministic, forward, single-product without routing (network equilibrium method).
- **Alizadeh et al. (2022)**: Multi-objective economic model, stochastic, closed-loop, multi-product (LP-Metric method).
- **Abdel-Azeem et al. (2022)**: Multi-objective economic model, deterministic, closed-loop, single-product (hybrid SLGP, LP-Metric, and SAW methods).
- **Nelson et al. (2022)**: Multi-objective model (economic-environmental), deterministic, closed-loop, multi-product (AEC method).
- **Jabbarzadeh et al. (2020)**: Multi-objective model (economic-environmental), deterministic, closed-loop, single-product (LP-Metric method).
- **Yavari & Zaker (2020)**: Multi-objective model (economic-environmental), forward, single-product incorporating vehicle routing (mixed bi-objective programming).
- **Onggo et al. (2019)**: Multi-objective economic model, stochastic, closed-loop, single-product with routing (heuristic-based methods and Monte Carlo simulation).

Table 2-2 Research Gaps

Row No.	Year	Authors	Objective Function					Parameter		Product		Supply Chain Type			Routing	Solution Method
			Economic	Social	Environmental	Multi-objective	Single-objective	Deterministic	Uncertain	Single	Multi	Direct	Reverse	Closed-loop		
1	2022	Jiang and colleagues	✓		✓		✓	✓		✓		✓				Network Equilibrium Method
2	2022	Alizadeh and colleagues	✓			✓			✓		✓			✓		LP-Metric
3	2022	Abdolazimi and colleagues	✓			✓		✓		✓				✓		SLGP, LP-Metric / SAW
4	2022	Nelson and colleagues	✓		✓	✓		✓			✓			✓		AEC
5	2020	Jabbari-zadeh and colleagues	✓		✓	✓		✓		✓				✓		LP-Metric
6	2020	Yavari & Shaker	✓		✓	✓		✓		✓		✓			✓	Bi-objective, Mixed-integer numerical programming
7	2019	Ounggo and colleagues	✓			✓			✓	✓				✓	✓	Heuristic, Monte Carlo Simulation
This Study			✓				✓		✓		✓			✓		single-objective optimization

Tabular overview of scholarly gaps across supply chain objectives and methods.

Research Questions

- **Primary Question:** How can a closed-loop supply chain model for perishable agricultural products be formulated to optimize inventory holding costs while minimizing waste generated by product decay?

Analytical Approach, Mathematical Formulation & Computational Framework

- **Network Structure:** A three-echelon supply chain comprising farms (suppliers), distribution/fulfillment centers, and recycling facilities/customers (retailers across Tehran's 22 municipal districts).
- **Model Assumptions:**
 - Retailer demand is predefined per time period based on regional population data.
 - Capacities for all distribution centers, recycling facilities, and transport vehicles are fixed.
 - Vehicle capacity, speed, and unit transit costs are specified.
 - Fixed costs for production, distribution, and recycling are predetermined.
 - Candidate locations for distribution and recycling centers are established.
 - Supply capacity limitations apply to each farm.
- **Mathematical Formulation (Objective Function & Constraints):**
 1. Objective Function: Minimizing total supply chain costs, including (i) fixed and variable routing/transportation expenses, (ii) inventory holding costs at distribution centers, (iii) stockout penalty costs, (iv) age-dependent product decay costs (b), and (v) reverse logistics and recycling operational expenses.
 2. Key Constraints: Vehicle and facility capacity restrictions, flow conservation at network nodes, subtour elimination constraints to guarantee route integrity, and product shelf-life thresholds.

Magnitudes $[S + W + G]$ with a fixed value at any given point are calculated and then added to the objective function.

$$Z = \min[R + N]$$

$$Z = \min(\sum_{p \in P} \sum_{\ell \in V} \sum_{j \in V} \sum_{k \in K} c_{ij} \cdot x_{ij}^{pk} + \sum_{i \in V} \sum_{j \in V} \sum_{p \in P} \sum_{k \in K} (\sum_A E_{ij}^{pkA} - \sum_F D_{ipo} \cdot x_{ij}^{pkF}))$$

Subject to:

a) In relation to calculating the costs of R .

$$R(X_{ij}^{pk}, D_{ip} | i, j \in V, p \in P, k \in K) = \sum_{p \in P} \sum_{\ell \in V} \sum_{j \in V} \sum_{k \in K} c_{ij} \cdot X_{ij}^{pk}$$

b) Also in relation to calculating the penalties of N .

$$N = \sum_{i \in V} \sum_{j \in V} \sum_{p \in P} \sum_{k \in K} (\sum_A E_{ij}^{pkA} - \sum_F D_{ipo} \cdot X_{ij}^{pkF})$$

c) As mentioned earlier, after calculating the minimum value of Z ($\min Z$), the final objective function M is defined, and the magnitudes with fixed value $[S + W + G]$ are added to it.

$$M = [Z + S + W + G]$$

$$M = [Z + \sum_{p \in P} \sum_{\ell \in V} s_{ip} + \sum_{p \in P} \sum_{\ell \in V} \sum_{b \in B} w_b \cdot l_{ipb} + \sum_{p \in P} \sum_{\ell \in V} g_{ip}]$$

d) Magnitude S , according to the above table, is the total overhead cost for each part i in the tour P .

$$S = \sum_{p \in P} \sum_{\ell \in V} s_{ip}$$

e) Magnitude W , according to the above table, is the total cost of corruption obtained.

$$W = \sum_{p \in P} \sum_{\ell \in V} \sum_{b \in B} w_b \cdot l_{ipb}$$

f) Magnitude G , according to the above table, is the total penalty cost of delay obtained.

$$g_{ip} \begin{cases} 0 & \text{if } d_{ipo} - l_{ipo} \leq 0 \\ 2 \cdot C_{i0} & \text{if } d_{ipo} - l_{ipo} > 0 \end{cases}$$

• Technical & Computational Tools:

- Coding and model formulation in GAMS.
- Execution using exact mathematical solvers (CPLEX / BARON).
- Implementation and validation using real-world distribution network datasets from the Tehran metropolitan area.

4. Key Results & Insights

Summary of Numerical Results & Sensitivity Analysis

- Empirical evaluation on the Tehran urban distribution network demonstrated high sensitivity of supply chain costs to seasonal demand shifts and maximum allowable shelf-life parameters (B).
- Sensitivity analysis under a $\pm 10\%$ demand variation confirmed linear behavior in total cost scaling relative to network workload changes.
- Extending the upper limit of product shelf life in colder seasons ($B=3 \rightarrow B=5$) successfully controlled degradation risks; however, it substantially increased overall holding costs due to higher inventory accumulation at distribution centers.

Managerial Insights

- Implementation of dynamic, season-adjusted warehousing and routing policies is critical, rather than relying on static yearly operational plans.
- Strategic resource allocation and capacity expansion should prioritize dense, high-demand areas (e.g., Districts 4, 5, and 15 of Tehran) to eliminate logistical bottlenecks.

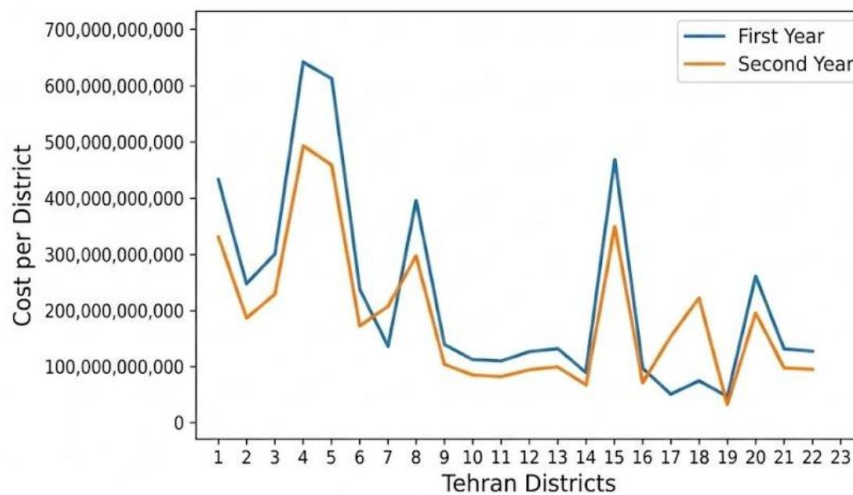


Figure 4-6 Comparison of supply chain cost per district over two consecutive years with different demand levels.

As illustrated in Figure 4-6, the implementation of the optimization model over two consecutive years with varying demand levels resulted in a significant reduction in total supply chain costs across most districts of Tehran (particularly Districts 4 and 5). Furthermore, the model's behavior in districts such as 17 and 18 demonstrates a smart reallocation of resources and facilities aimed at minimizing total network costs in response to demand fluctuations.

The calculated supply chain costs are measured and reported in Iranian Rials (IRR) per kilogram of perishable agricultural products (IRR/kg).

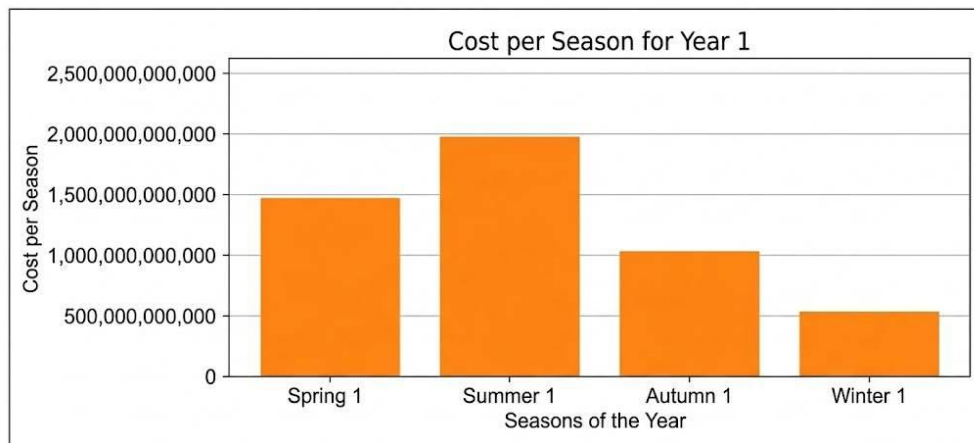


Chart 4-2 Seasonal cost based on first year demands
Comparison of cost variation across four quarters

In Chart 4-2, the supply chain costs in the first year exhibit a distinct seasonal pattern. The total cost reaches its peak during the summer season, primarily driven by higher ambient temperatures, increased decay rates of perishable agricultural products, and the resulting rise in cold-chain storage and transportation expenses. Conversely, with the drop in temperature and shifts in seasonal demand during autumn and winter, supply chain costs decrease significantly. This seasonal insight enables decision-makers to optimize resource allocation and operational budgeting according to peak demand periods.

Final cost comparison over two consecutive years with different demand levels

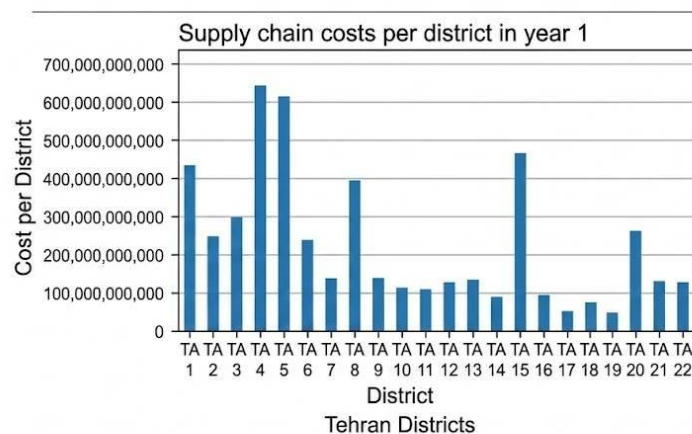


Figure 4-3 Supply chain cost per district with first year demand
Bar chart indicating highly skewed cost distribution across districts

5. Research Contributions & Associated Outputs

- **Development of a Dynamic Multi-Period Mathematical Model:** Incorporating the simultaneous impacts of seasonal temperature variations on product decay rates and demand fluctuations over a multi-year horizon.
- **Spatial Cost & Hotspot Identification:** Uncovering the highly skewed cost distribution across Tehran's 22 districts and pinpointing high-cost nodes (e.g., Districts 4, 5, and 15) to guide targeted cold-chain investments and budget control.
- **Analysis of Seasonal Supply Chain Dynamics:** Demonstrating cost peaks during the summer season (driven by ambient temperatures and refrigeration needs) and providing a financial blueprint for operational cash-flow management across all four quarters.
- **Multi-Year Model Validation & Adaptability:** Proving the model's ability to efficiently reallocate resources and achieve significant cost reductions in subsequent years under varying demand scenarios.

6. Alignment with Proposed PhD Research
